ANGSHUMAN MAZUMDAR

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SKILLS

Programming Languages: C++, C#, CUDA, OpenGL, Vulkan, VBA, Python, HTML
Engines & Frameworks: Unity (Desktop & VR), Unreal Engine (Blueprint & C++), Raylib
Simulation: NetLogo, AnyLogic, Houdini
Modelling: Siemens NX, Solidworks, Blender
Data & Workflow Tools: Teamcenter PLM, MS Power Automate, Git, Jira, Qualtrics, NXOpen, Solidworks API
Documentation & Technical Communication: Doxygen, Overleaf, Markdown, Confluence
Soft Skills: Mentorship, Leadership, Teaching

PROFESSIONAL EXPERIENCE

Games Innovation Laboratory – Purdue University (08/2021 – Present)
Graduate Mentor and Research Advisor

- Guided multidisciplinary teams in VR development, simulation design, and real-time graphics research.
- Advised projects involving agent AI, optimization algorithms, and immersive experience design.

School of Engineering Technology – Purdue University (08/2020 – Present)
Lead Graduate Teaching Assistant (NX, Aras, & Teamcenter)

- Led a team of 10 TAs supporting 1000+ students across engineering graphics and PLM courses.
- Built Unity, VBA, and Power Automate-based tools to reduce grading time by 50%.
- Designed frameworks for continuous improvement (CM2) in engineering graphics education.
- Recognized with the **Graduate Teaching Award** by the Purdue University Teaching Academy

Bharti Airtel (05/2018 - 06/2018)
Summer Internship (Networking Department)

- Developed automated workflows for data collection and validation.
- Built VBA tools to reduce manual processing and improve reliability.

EDUCATION

Ph.D. in Technology (2021 - 2026)
Purdue University

M.S. in Computer Graphics Technology (2019 - 2021)
Purdue University

B. Tech. in Electronics and Communication Engineering (2015 - 2019)
Vellore Institute of Technology

GAME PROJECTS

Vasoo Virtual Stadium Experience – *Interaction Mechanic Developer (Iconic Engine)* (2022): Implemented core interaction systems for large-scale virtual stadium environments.

Sonata Theory (Steam) – *Audio Producer (MAYJ Studios)* (2021): Contributed to audio systems and production pipeline for a shipped indie title.

PUBLICATION & RESEARCH EXPERIENCE

Enterprise PDM as Digital Backbone in a Large First-Year Engineering Course: Formalized PLM/PDM and CM2-based curricula for engineering education and digital enterprise integration. (ASEE 2025, Canada)

Engineering Graphics Education for the Digital Enterprise: Introduced project-based learning and continuous improvement frameworks in engineering graphics courses. (ASEE 2024, Portland, OR)

Synthesizing Affective Virtual Reality Multicharacter Experiences (Master's Thesis): Developed optimization algorithms using Markov Chain Monte Carlo and Simulated Annealing for affective VR experiences. (CASA 2021, Canada)

Virtual reality game level layout design for real environment constraints: Built ECS based intelligent patrol agents with zone-based detection, visibility-cost modeling, and follow intelligence for Oculus Quest. (Conference of CAD & Graphics 2021, China)

Embodiment for the Difference: A VR Experience of Bipolar Disorder: Developed an immersive simulation embodying bipolar disorder for the 3DUI Contest. (IEEEVR 2020, Atlanta, GA)

Walking in a Crowd Full of Virtual Characters: Effects of Virtual Character Appearance on Human Movement Behavior: Created pipelines for optimizing Mixamo assets and integrating them into Unity for crowd simulation studies. (ISVC 2020, San Diego, CA)